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## DESIGN AND DEVELOPMENT OF FUNCTIONAL MODULES FOR A LEARNING MANAGEMENT SYSTEM

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# DESIGN AND DEVELOPMENT OF FUNCTIONAL MODULES FOR A LEARNING MANAGEMENT SYSTEM

## ABSTRACT

Learning Management Systems (LMS) are essential platforms in educational institutions for managing academic operations including attendance tracking, examination administration, and stakeholder communication. As these systems grow in complexity, monolithic architectures become bottlenecks in scalability and maintainability. This project focuses on improving selected modules of an existing LMS used at Codegnan IT Solutions Pvt Ltd. Instead of maintaining the earlier tightly connected backend, the chosen modules were redesigned to function as separate services. The work includes redevelopment and enhancement of three main modules: an exam management module, an exam reporting module, and a parent reporting module. The exam module supports question handling, batch-level exam tracking, and student notifications. The reporting module provides exam-related summaries and performance information. The parent reporting module generates student performance reports, prepares PDF files, and shares them through WhatsApp. Report generation is handled separately so that it does not interrupt other LMS activities. The modules were developed using Python and MongoDB and organized in a way that allows them to operate independently. This approach reduces strong dependencies between components and simplifies future updates. During the transition from the earlier backend structure, unnecessary elements were removed to keep the system simpler and easier to manage. The project was carried out between December 2025 and March 2026 as part of the LMS redevelopment process at Codegnan IT Solutions Pvt Ltd.

**Keywords:** Learning Management System, Microservices, Flask, MongoDB, REST API, CQRS, WhatsApp Integration, PDF Generation, AWS, Celery, Redis